

BASE MARAMBIO, Antarctica

WMO: 890550

Lat: 64.241S

Lon: 56.625W

Elev: 656

StdP: 14.35

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-21.2	-17.7	-26.9	1.3	-19.8	-23.1	1.6	-16.9	68.4	-1.4	62.6	-7.7	30.3	200	1.259

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.6	45.5	38.9	42.9	37.2	40.6	35.5	39.5	44.3	37.8	41.9	36.2	39.7	10.9	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
35.8	31.5	39.7	34.2	29.6	37.7	32.9	28.1	36.3	15.3	44.6	14.5	42.1	13.8	40.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 56.4	50.1	44.8	-23.6	50.9	4.7	3.4	-26.9	53.4	-29.7	55.4	-32.3	57.3	-35.7	59.8		
(5)			-23.7	44.0	3.9	2.8	-26.4	46.0	-28.7	47.7	-30.9	49.3	-33.7	51.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	30.6	29.0	22.7	13.2	11.4	5.3	4.9	7.9	12.7	19.2	25.5	29.5
	DBStd	14.58	4.28	6.39	9.82	12.87	13.42	14.28	14.74	13.95	14.41	11.54	7.17	4.54
	HDD50	11831	603	588	848	1105	1198	1340	1399	1305	1119	956	736	634
	HDD65	17306	1068	1008	1313	1555	1663	1790	1864	1770	1569	1421	1186	1099
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	18.7	14.2	15.9	18.1	19.5	18.9	21.2	20.7	20.7	21.1	19.9	18.3	16.3
Precipitation	PrecAvg	33.5	2.5	2.8	3.6	3.0	2.7	2.1	2.4	2.6	3.0	3.5	3.1	2.3
	PrecMax	43.5	3.6	4.0	4.4	5.2	4.6	3.7	4.2	5.1	4.7	5.3	4.8	3.9
	PrecMin	23.1	1.5	1.8	2.0	0.8	1.2	0.8	0.6	0.4	1.5	2.3	1.5	0.9
	PrecStd	6.0	0.6	0.6	0.6	1.0	0.8	0.7	1.0	1.1	1.0	0.8	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.0	48.6	45.5	43.0	46.4	39.1	40.0	38.1	41.6	43.0	47.4	45.2
		MCWB	42.2	41.2	39.2	36.0	38.5	31.9	32.9	32.4	34.5	36.3	40.4	38.7
	2%	DB	43.8	44.5	41.9	37.8	39.7	34.7	34.1	33.9	37.6	39.0	41.6	41.4
		MCWB	38.6	38.3	36.6	33.3	34.2	31.2	30.5	30.1	33.0	34.1	36.5	36.2
	5%	DB	40.4	41.4	38.9	35.0	35.7	31.6	30.7	30.9	35.1	36.7	38.0	38.9
		MCWB	36.2	36.4	34.7	32.1	31.8	29.6	28.8	28.9	31.9	32.8	34.1	34.7
	10%	DB	37.4	38.7	35.9	32.2	31.9	28.6	28.2	28.4	32.6	34.5	35.3	36.6
		MCWB	34.4	34.8	33.2	30.4	30.2	27.2	26.8	27.0	30.5	31.7	32.5	33.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	42.7	41.8	40.2	36.9	38.8	33.3	33.7	33.6	36.0	37.4	40.9	39.5
		MCDB	46.7	47.6	44.3	41.6	45.7	35.8	37.7	36.5	39.5	41.7	47.5	44.0
	2%	WB	38.9	39.0	37.2	34.1	34.8	31.5	31.1	30.8	33.6	34.8	37.1	36.9
		MCDB	43.0	43.5	41.1	36.9	38.9	34.3	33.6	32.9	36.5	37.9	41.2	40.4
	5%	WB	36.7	36.8	35.1	32.2	32.2	29.6	29.0	29.0	32.1	33.2	34.4	35.3
		MCDB	39.8	40.8	38.2	34.8	34.9	31.5	30.7	30.7	34.5	36.0	37.4	38.2
	10%	WB	34.7	35.0	33.3	30.2	30.1	27.1	26.8	27.0	30.5	31.9	32.6	33.6
		MCDB	37.2	38.3	35.7	32.3	32.0	28.8	28.3	28.7	32.7	34.3	34.9	36.2
Mean Daily Temperature Range	5% DB	MDBR	6.6	7.7	9.5	11.2	12.7	11.5	13.6	13.2	13.1	10.7	8.2	6.7
		MCDBR	11.7	12.6	12.3	14.2	13.3	13.8	15.3	15.1	13.6	11.8	11.9	10.8
		MCWBR	8.1	8.8	9.2	11.3	9.5	12.0	13.5	13.1	10.7	8.8	8.7	7.4
	5% WB	MCDBR	11.3	12.1	11.6	14.4	13.5	13.9	15.3	15.1	12.7	11.3	11.2	10.3
		MCWBR	8.3	9.0	9.3	12.6	11.1	12.6	14.0	13.8	10.9	9.1	8.6	7.6
Clear-Sky Solar Irradiance	taub	0.278	0.272	0.258	0.225	0.152	0.139	0.142	0.213	0.244	0.262	0.266	0.275	
	taud	2.440	2.453	2.454	2.331	1.993	1.933	1.956	2.243	2.306	2.326	2.375	2.373	
	Ebn at Noon	315	301	277	227	148	70	146	225	274	300	317	320	
	Edh at Noon	32	28	22	15	7	3	7	15	24	31	34	35	
All-Sky Solar Radiation	RadAvg	1985	1417	798	328	78	18	47	233	708	1380	2020	2294	
	RadStd	202	132	98	29	5	1	2	17	37	54	102	227	

Historical Trends

		Heating		Cooling		Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page